MIM-HeMIS: Self-Supervised Masked Image Modeling with Heterogeneous Modality Fusion for Brain Tumor Segmentation under Missing MRI Modalities

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***Abstract*—Accurate segmentation of regions of interest from MRI to diagnose brain tumors is crucial. MRI is the standard technique used in imaging the brain, but manual segmentation of tumor regions is time-consuming, manual and prone to human error. Although deep learning has been able to significantly enhance segmentation performance, it requires big data sets of labeled images which is difficult in medical applications. In this study, we introduce a hybrid CNN-Transformer framework that integrates the Masked Image Modeling (MIM) based self-supervised pretraining and the HeMIS statistical abstraction mechanism. A hybrid model is pretrained on the BraTS 2021 dataset and finetuned for 4-class brain tumor segmentation for all 4 MRI modalities following the proposed framework. A major advantage is that the modalities are robust, meaning that there is no need to train a new model for each of the 15 possible combinations of modalities. When modalities are missing, the variance computed by the HeMIS abstraction layer captures feature disagreement across available inputs, providing useful information for identifying cases where predictions may be less reliable. Though this variance has not been calibrated as a clinical confidence score. The model was tested on BraTS 2021 with all 15 modality combinations with excellent and clinically applicable results: Whole Tumor Dice of 90.2%, Tumor Core Dice of 85.9%, and Enhancing Tumor Dice of 79.1% for all four modalities.**

***Index Terms*—Brain tumor segmentation, BraTS 2021, self-supervised learning, masked image modeling, missing MRI modalities, HeMIS, multimodal MRI, 3D segmentation.**

I. Introduction

Brain tumor segmentation from magnetic resonance imaging (MRI) is one of the core problems in medical image analysis, as it is vital for the localization of abnormal tissue, estimation of tumor burden, treatment planning and longitudinal monitoring. Standard radiological evaluation uses various MRI sequences to complement each other in order to give clinical information. FLAIR is useful for edema and abnormalities that are hyperintense, T1ce is useful for displaying lesions with enhancement, T2 is useful for identifying lesions that are fluid sensitive, and T1 highlights anatomic structure. Thus, a segmentation system with these modalities can take advantage of both anatomical and pathological information. This is a very important task but manual contouring is still a very labor intensive process and relies on expert radiologists.

Brain tumors are also quite different in size, location, strength and subregion-dominance. These factors drive up the cost of voxel-level annotation, and the development of a supervised model is difficult in the presence of few annotated cases. In addition, research datasets may contain all four MRI modalities, but not necessarily clinical datasets. Modalities may be missing due to time limits during acquisition, patients condition, cost, availability of the scanner, protocol differences, or imaging artifacts. A model that fails if one of the input channels is missing is not as useful in such environments. This work tackles two closely related problems: reliance on labeled segmentation information and sensitivity to missing modalities.

The proposed approach is an unsupervised volumetric MRI representation learning via masked image modeling, followed by supervised fine-tuning. It also adopts the feature abstraction concept of HeMIS, if there is any, by fusing the features from different modalities, across them, by mean and variance statistics, instead of assuming that the full set of input features is available. The resulting architecture will allow for the use of a single trained model for all non-empty subsets of the 4 MRI modalities. The major contributions are outlined below. The first part of the paper is the introduction of a hybrid SSL+HeMIS architecture for 3D brain tumor segmentation in the context of incomplete modality. Second, it learns MRI patch representations with the aid of MIM pretraining without segmentation labels. Thirdly, it tested the trained model on all 15 non-empty combinations of FLAIR, T1ce, T1 and T2. Fourth, it shows the behavior of the proposed model in comparison with SimCLR and Med3D baselines, and examines modality-count, modality-identity and HeMIS-variance behavior.

The work is not meant to be a clinical diagnostic system by itself; it is presented as a research level segmentation framework. The research goals that will be carried over to the technical objectives are to develop a 3D segmentation model for encoder-decoder, incorporate the four MRI modalities as complementary features, implement modality-aware fusion for handling modalities with missing sequences, and evaluate clinically relevant regions such as Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

II. Related Work

Deep learning for brain tumor segmentation has come a long way from convolutional encoder-decoder networks to self-supervised and multimodal approaches. U-Net-like models are effective because they preserve spatial localization and skip connections. However, many segmentation pipelines still require complete MRI inputs and will degrade its performance when one sequence is unavailable.

The idea of Hetero-Modal Image Segmentation (HeMIS) was introduced in [1] which was directly related to missing-modality segmentation, where modality features were summarized by using statistical abstraction instead of a fixed channel configuration. This concept is especially true of MRI as all sequences differ in their contrast properties. The current framework takes a similar approach by computing mean and variance feature maps by available modalities encoders.

The other important direction is self-supervised learning (SSL) as it is very expensive to have expert labels for medical imaging. The contrastive learning methods like SSCLNet and SimCLR style learn the representation of features through comparison of different views of images [2], [13], [14]. Generative and reconstruction-based approaches, such as masked autoencoders and masked image modeling, train by predicting the content of missing parts of an image or reconstructing the image from a set of corrupted parts. Recently, several studies explored the application of SSL to MRI segmentation and classification such as student-teacher learning [3] and enhanced SSL with multimodal MRI [6] and large-scale medical vision models [11], [20]. The motivation of SSL is the problem of the scarcity of labeled medical data, which has been repeatedly stated in the literature.

Masked image modeling is particularly relevant to volumetric MRI. There have been anatomical prior-informed masked autoencoders, which are used to guide the pretraining process [5]; there have been cross-contrast or multimodal masked autoencoder models for brain MRI for brain tumor analysis [9], [18]. The results of these studies demonstrate that reconstruction-based pretraining can be used to acquire useful spatial and cross-modal representations. The objective function is proposed to be MIM, but it is fused with HeMIS style fusion instead of just a Transformer Encoder.

Other works are devoted to architectural enhancements or segmentation robustness. A variety of current research trends can be seen in residual vision transformers, wavelet attention, edge-aware SSL, efficient U-Net variants, diffusion-based MRI synthesis, and wide-ranging reviews of deep learning for brain tumor segmentation, among others [7], [8], [10], [12], [15]–[17], [19]. The BraTS-METS challenge additionally underscores the significance of dealing with the variability of lesions and the heterogeneity of data that segmentation systems have to address [4].

III. Design Requirements and Scope

The proposed model is a 3D brain MRI research-level hybrid segmentation model. The model should be able to receive 3D MRI volumes, process multiple MRI modalities, produce a segmentation mask of the tumor, allow SSL pretraining and be tested on the missing modalities condition.

The proposed model is expected to work even when only one modality is available and also when there are different combinations of modalities. The final assessment is being carried out on each of the 15 nonempty subsets of the four MRI modalities. The non-functional requirements focus on reproducibility, Dice-based performance assessment, documented training environments and the feasibility of execution on the available GPU environment. Practical limitations included in the research are: limited to BraTS 2021, no direct deployment in the hospital, influenced by GPUs memory, preprocessing options, training time, and availability of modalities. These constraints are retained to avoid overstating the model beyond experimental setting.

The alternative design directions that were considered in the research are summarized in Table I. SimCLR can serve as a 3D self-supervised contrastive baseline and Med3D can serve as a supervised volumetric medical-imaging baseline. However both of these do not have an explicit mechanism for the availability of variable modalities. The selection of SSL+HeMIS is made due to the fusion of direct available-modality and representation learning.

TABLE I
DESIGN-LEVEL COMPARISON OF CONSIDERED SOLUTIONS.

Design	Strength	Limitation / decision
SimCLR	Learns SSL-based 3D visual representations	No direct missing-modality fusion; used as baseline
Med3D	Processes volumetric medical images and 3D spatial features	Assumes stable complete-input conditions; used as baseline
SSL+HeMIS	Combines MIM pretraining with HeMIS-style fusion	More complex, but selected for missing-modality robustness

IV. Materials and Methods

A. Dataset

The data to be used in the experiments is the BraTS 2021 dataset. The dataset includes 1,251 samples with each having four MRI modalities. A voxel-wise segmentation mask is also present in each of the cases. They are background and healthy tissues, necrotic and non-enhancing tumor core (NCR/NET), peritumoral edema (ED), for enhancing tumor (ET) . The original BraTS tumor label 4 is remapped to become tumor label 3 in such a way that the class index set is consecutive: 0, 1, 2, and 3. The data is separated into five folds. Model training is performed on Folds 1–4 that include 1,000 subjects, whereas the model is evaluated on Fold 0 which includes 251 subjects. This subject-level isolation occurs prior to preprocessing, SSL pretraining, supervised fine-tuning, and final evaluation, so that validation subjects are not used to train the model or to assess its normality. Data The tumor load of the dataset is heterogeneous. The study demonstrates an average tumor volume of 96.0 cm^3 and a central tumor volume of 89.3 cm^3 . The subjects can also be classified as small, medium, large and very large tumors. This increases the difficulty of segmentation since the model will need to deal with the triviality of small and large tumor extents.

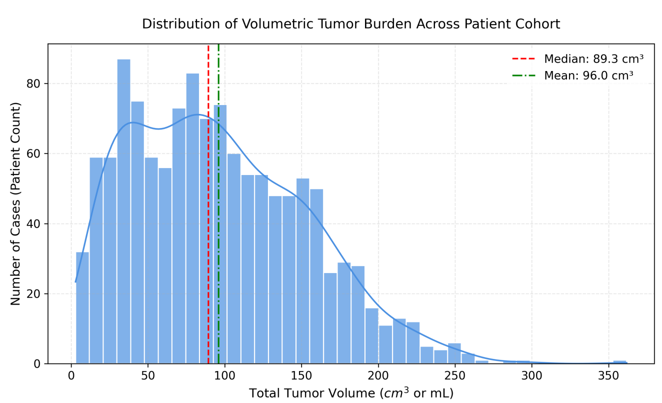

Fig. 1. Tumor-volume distribution across the BraTS 2021 cohort used in the thesis. The distribution shows substantial variation in tumor burden, with mean volume 96.0 cm^3 and median volume 89.3 cm^3.

B. Preprocessing

Prior to training, all modalities are standardized. To keep the memory consumption low and the input to MRI volumes fixed, the volumes are cropped in the center to a size of $128 \times 128 \times 128$ voxels. Normalization is done independently on each modality in the non zero foreground voxels. First it uses percentile clipping at the 0.05^{th} and 99.95^{th} percentile to remove extreme intensity outliers.

$$X_m^{norm}(i) = \frac{X_m^{clip}(i) - \mu_m}{\sigma_m + 10^{-8}}, \tag{1}$$

where m denotes a modality, i denotes a voxel, and μ_m and σ_m are computed only from non-zero voxels in that subject and modality. Training augmentation consists of random flips along the 3D spatial axis, brightness jitter and contrast jitter. Validation and evaluation use deterministic preprocessing.

C. Overall Framework

There are two stages to the proposed workflow. The model is pretrained with the masked image modeling in Stage 1. Four modalities of MRI are concatenated and split into 3D patches, with some of the patches being masked. The masked patch content is reconstructed in the SSL branch with the loss function of MSE. This stage is training the representation without the segmentation labels. In Stage 2, the pretrained representation is used in the segmentation model and fine-tuned using the four-class voxel-wise tumor segmentation label.

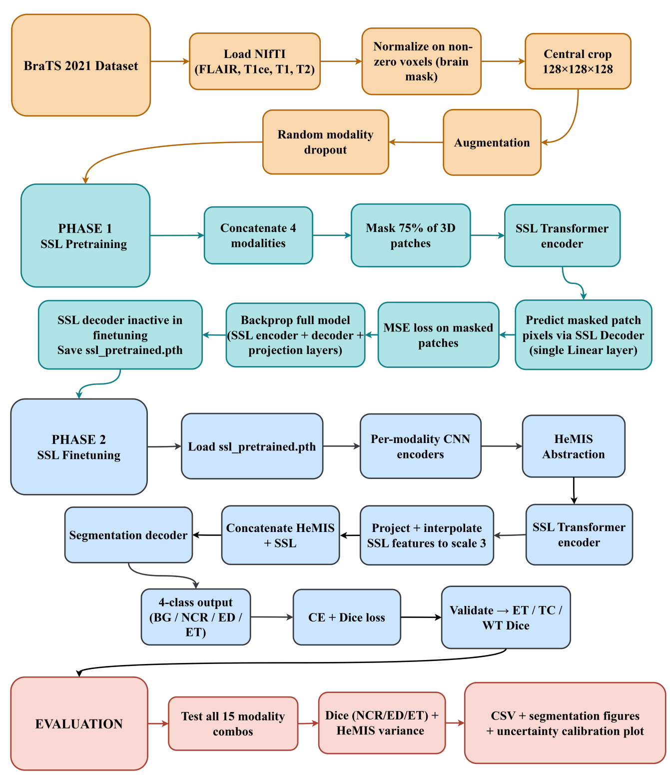

Fig. 2. Workflow of the proposed MIM-HeMIS framework.

The framework has two phases. The first phase is used for multiscale feature maps with per-modality CNN encoders. Multiscale feature maps from per-modality CNN encoders in the first phase. The second phase is based on an SSL Transformer encoder to learn global patch-level MRI representations. The HeMIS abstraction combines elements of modalities currently present, and the HeMIS bottleneck is followed by the projected SSL features during decoding.

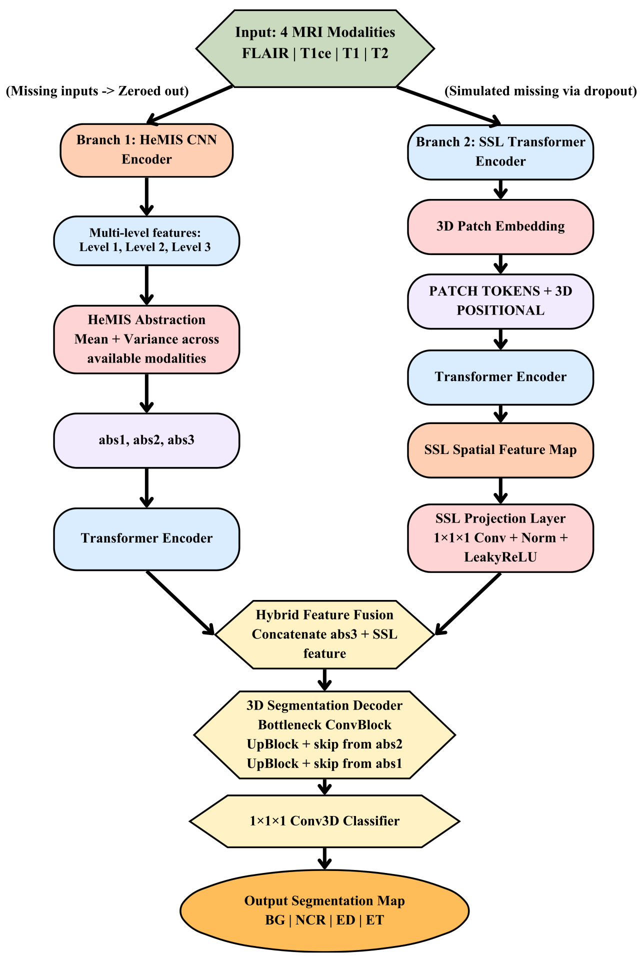

Fig. 3. Hybrid HeMIS-SSL architecture.

D. HeMIS-Based Feature Fusion

For each available modality, a CNN encoder extracts features at multiple scales. At a given scale s, the set of available modality feature maps is denoted by $\mathcal{M}$. The HeMIS abstraction computes the element-wise mean and variance:

$$\mu^s = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} f_m^s, \tag{2}$$

$$(\sigma^s)^2 = \frac{1}{|\mathcal{M}|} \sum_{m \in \mathcal{M}} (f_m^s - \mu^s)^2, \tag{3}$$

$$F_{\text{HeMIS}}^s = \text{Concat}(\mu^s, (\sigma^s)^2). \tag{4}$$

Here, f_m^s is the feature map extracted from modality m at scale s. Missing modalities are excluded from the computation. If only one modality is available, the variance tensor is set to zero. This allows a single model to process any non-empty modality subset.

CNN has three base channels: 32, 64 and 128. The fused HeMIS features are used as decoder skip connections, and the depth of the HeMIS feature is fused with the projected SSL features at the bottleneck. The 3D U-Net style compact pattern is used for each modality encoder. The first level of the encoders runs at full $128 \times 128 \times 128$ resolution, the second level of the encoders runs after one max-pooling downsampling step, and the third level is after another max-pooling downsampling step. Each convolutional block is made of two Instance Normalization and Leaky ReLU layers of convolution, each performing $3 \times 3 \times 3$ convolution. It is done using Instance Normalization because the fine-tuning batch size is one, making batchdependent normalization unstable for 3D medical volumes. The research is based on a 640 channel bottleneck made up of 256 HeMIS channels and 384 SSL channels. Two upsampling stages with skip connections from shallower HeMIS abstraction layers are then executed by the decoder. The last $1 \times 1 \times 1$ 3D convolution results in 4 classes for background, NCR, ED and ET.

E. Masked Image Modeling Pretraining

The four MRI modalities are considered as multi-channel 3D input in the SSL branch. A patch embedding layer divides a $128 \times 128 \times 128$ volume into non-overlapping $16 \times 16 \times 16$ patches, giving $8 \times 8 \times 8 = 512$ patch tokens. A projection into an embedding dimension of 384 is made for each token. 3D sinusoidal positional encoding is used to maintain spatiotemporal position in depth, height and width. The Transformer encoder consists of 4 layers and 8 attention heads.

The pretraining objective is masked patch reconstruction. For a masked patch x_i, the target is normalized per patch and the reconstruction error is computed only at masked positions:

$$\hat{x}_i = \frac{x_i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}, \tag{5}$$

$$\mathcal{L}_{\text{MIM}} = \frac{1}{|\mathcal{M}_{mask}|} \sum_{i \in \mathcal{M}_{mask}} \|\tilde{x}_i - \hat{x}_i\|_2^2. \tag{6}$$

The mask ratio that is displayed in the final configuration table in the research is 0.75. The learning rate is 1×10^{-5} and weight decay 0.05, while the rest of the training is done in SSL. The pretrained checkpoint is then used to initialize the segmentation model. The training design also features several implementation safeguards. Tensors of the preprocessed data are cached to prevent repeated loading of NIfTI data, and DataLoader options like pinned memory, prefetching, persistent workers, and non-blocking data transfers are employed to minimize GPU idle time. The batches that contain NaN or Inf loss values are ignored during training and gradients are clipped to avoid unstable updates. Choices are significant due to the input being a 3D volume with high memory and the high cost of computation of SSL pretraining and segmentation fine-tuning.

F. Supervised Fine-Tuning and Missing-Modality Simulation

In supervised fine tuning, segmentation labels are provided, and modalities are randomly dropped with 30% probability to simulate the unavailable MRI modality. There is a safety rule that guarantees a modality is available. In case of a missing modality, a channel placeholder with a zero channel is used in the SSL branch and mean and variance is calculated only based on existing modality features in the HeMIS branch. This maintains one architecture for all combinations of modalities.

The segmentation loss combines weighted Cross-Entropy and Dice loss with equal weights:

$$\mathcal{L}_{seg} = 0.5\mathcal{L}_{CE} + 0.5\mathcal{L}_{Dice}. \tag{7}$$

The Cross-Entropy class weights are 0.1 for background, 2.0 for NCR, 1.5 for ED, and 2.0 for ET. AdamW is used with learning rate 2×10^{-4} and weight decay 0.05. Fine-tuning uses mixed precision, gradient clipping with maximum norm 1.0, and checkpointing according to the highest validation Dice.

TABLE II
KEY CONFIGURATION OF THE PROPOSED HEMIS-SSL MODEL.

Parameter	Value
Dataset	BraTS 2021, 1,251 cases
Train / validation split	1,000 / 251
Modalities	FLAIR, T1ce, T1, T2
Input size	$128 \times 128 \times 128$ voxels
SSL patch size	$16 \times 16 \times 16$
Number of patches	512
SSL embedding dimension	384
Transformer depth / heads	4 / 8
Mask ratio	0.75
Missing modality probability	0.30 during fine-tuning
Base CNN channels	32
Segmentation classes	BG, NCR, ED, ET
Pretrain optimizer	AdamW, lr=1×10^{-5}
Fine-tune optimizer	AdamW, lr=2×10^{-5}
CE / Dice loss weight	0.5 / 0.5

G. Baseline Designs

To provide a comparison for the research, two baseline directions are kept. The first is a baseline of a self-supervised model using 3D SimCLR style. It uses a 3D ResNet style encoder with four input channels: FLAIR, T1ce, T1, T2. In the pre-training stage, two views of the same MRI volume are learnt to be closer in representation space while different subjects are learnt to be farther away from each other through contrastive learning. In the fine tuning stage, an encoder is paired with a decoder similar to a U-Net. The research reveals an all four modality Dice value of 0.724, and an average Dice across all 15 combinations of 0.368.

The second baseline model is a supervised 3D model similar to Med3D. It learns the four modalities as a volumetric tensor and applies 3D convolutional residual blocks to capture spatial knowledge along depth, height and width. The first convolution blocks are fine-tuned for 4 input channels, and the last layer outputs four classes of segmentation. This baseline is helpful to evaluate the performance of 3D supervised segmentation but is not considering SSL pretraining or HeMIS fusion. The research results are reported as 0.704 Dice (all four modalities) and 0.335 Dice (average across all modality combinations).

The proposed SSL+HeMIS model is different from both baselines as it explicitly models variable input availability. It does not create artificial missing sequences, either. In lieu of the reconstruction of missing modalities as images, it directly combines the characteristics of modalities that are present. This design option is easier to deploy as there is no need for an additional modality-synthesis step and because it does not display a generated MRI image as a clinical image.

H. Evaluation Metrics

Evaluation is carried out for all 15 non-empty subsets of the four MRI modalities. NCR, ED and ET are evaluated by labelwise dice scores. The label mean Dice is the average of these three tumor labels. The BraTS-style clinical regions are also assessed: ET = label 3, Tumor Core (TC) = NCR + ET, Whole Tumor (WT) = TC + all tumor labels. Dice overlap is defined as:

$$\mathrm{Dice}(P, T) = \frac{2|P \cap T|}{|P| + |T| + \epsilon}, \tag{8}$$

where P is the predicted tumor region and T is the ground-truth region. Precision, recall, F1-score, correlation statistics, and qualitative segmentation visualizations are also reported in the thesis.

V. RESULTS

A. Performance Across Modality Combinations

The comparison between SimCLR, Med3D, and SSL+HeMIS for all modality combinations except empty is shown in Table III. The proposed SSL+HeMIS model achieves the maximum Dice score in all the combinations listed. The best results are obtained for the full four modality setting where the mean Dice is equal to 0.766. The setup of the [FLAIR+T1ce+T2] gives a close to full input combination of 0.763, indicating that the model is still robust without the T1 input.

For all four modalities, ET is 0.791, TC is 0.859 and WT is 0.902 for the clinical region Dice score. The clinical mean Dice is 0.851. The results of these scores indicate that a high performance is achieved if maximum complementary information from all MRI sequences can be used. The FLAIR+T1ce+T2 combination works well too, only slightly less with mean Dice 0.763 and clinical mean Dice 0.847.

For single modalities, T1ce has the highest input with a Dice score of 0.532 for the label and 0.611 for the clinical. FLAIR and T2 also provide useful single-modality performance due to the ability to highlight edema and abnormal tissue. The anatomical structure alone is not enough for detailed segmentation of the tumor subregions as evidenced by the lowest modality T1 with a label mean Dice 0.214.

TABLE III
COMPARISON OF SIMCLR, MED3D, AND SSL+HEMIS ACROSS 15 MODALITY COMBINATIONS. VALUES ARE DICE SCORES REPORTED IN THE RESEARCH.

Combination	SimCLR	Med3D	SSL+HeMIS	Combination	SimCLR	Med3D	SSL+HeMIS
F	0.235	0.2017	0.418	T1c + T2	0.421	0.4180	0.719
T1c	0.298	0.0718	0.532	T1 + T2	0.013	0.1276	0.468
T1	0.001	0.0121	0.214	F + T1c + T1	0.706	0.6468	0.754
T2	0.060	0.1102	0.424	F + T1c + T2	0.681	0.5582	0.763
F + T1c	0.622	0.4612	0.745	F + T1 + T2	0.334	0.3215	0.551
F + T1	0.306	0.2689	0.486	T1c + T1 + T2	0.450	0.5918	0.732
F + T2	0.288	0.2845	0.539	All 4	0.724	0.7044	0.766
T1c + T1	0.376	0.2437	0.654				

B. Effect of Modality Count

Table IV gives the average performance for each number of modalities available. Label mean Dice goes up from 0.397 with 1 modality to 0.602 with 2 modalities to 0.700 with 3 modalities to 0.766 with all 4 modalities. Similar mean Dice results are observed from 0.517 to 0.851. The pattern establishes the importance of multimodal MRI for segmentation.

TABLE IV
AVERAGE PERFORMANCE BY NUMBER OF AVAILABLE MODALITIES.

Available modalities	Label mean Dice	Clinical mean Dice
1 modality	0.397	0.517
2 modalities	0.602	0.719
3 modalities	0.700	0.798
4 modalities	0.766	0.851

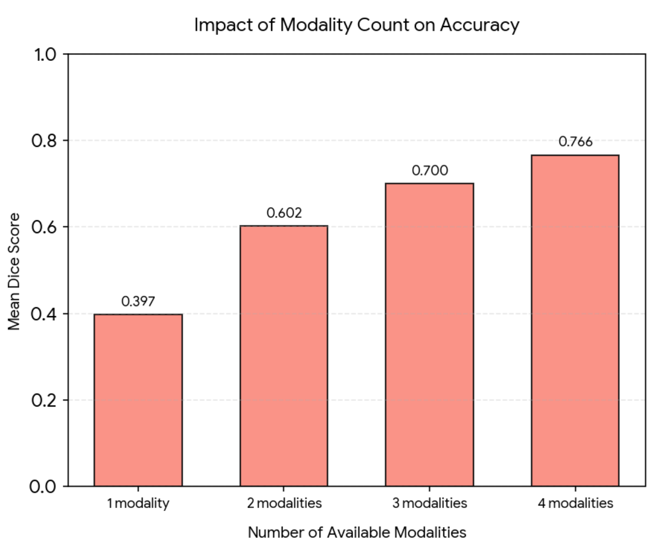

Fig. 4. Impact of modality count on mean Dice score. The average Dice increases as more MRI modalities are available.

The results also indicate that the modality identity is important, however. Modality count is not sufficient for performance. Combinations which include T1ce tend to work well because they include contrast enhanced information for both ET and TC. FLAIR and T2 are useful for edema, whole tumor borders. This is why FLAIR+T1ce+T2 is close to the all four modality results.

The descriptive statistical summary by modality is reported with Table V. The mean values in each case increase, illustrating that the more modalities, the better the segmentation, and the standard deviations within the one-, two-, and three modality groups illustrate that the precise subset of modalities is also important.

TABLE V
STATISTICAL SUMMARY BY NUMBER OF AVAILABLE MODALITIES.

Available modalities	Label Dice	Clinical Dice	Macro F1
1 modality	0.397 ± 0.133	0.516 ± 0.130	0.464 ± 0.133
2 modalities	0.602 ± 0.120	0.719 ± 0.090	0.669 ± 0.119
3 modalities	0.700 ± 0.101	0.798 ± 0.075	0.768 ± 0.103
4 modalities	0.766	0.851	0.831

C. Aggregate Baseline Comparison

The overall comparison of the final design candidates is shown in Table VI. Overall, across the 15 modality combinations, SSL+HeMIS achieves a mean Dice of 0.584, while SimCLR and Med3D achieve a mean Dice of 0.368 and 0.335, respectively. It also offers the highest of all 4 modality Dice. This validates the choice of SSL+HeMIS as the ultimate model as it is more robust in the complete-input scenario as well as in missing-modality scenarios.

TABLE VI
AGGREGATE PERFORMANCE COMPARISON OF FINAL DESIGN CANDIDATES.

Model	Mean Dice across 15	All 4 Dice	Status
SimCLR	0.368	0.724	Baseline
Med3D	0.335	0.704	Baseline
SSL+HeMIS	0.584	0.766	Selected

As demonstrated by the configuration-level paired comparison in this research, SSL+HeMIS achieves a mean Dice improvement of 0.217 over SimCLR and 0.250 over Med3D when considering the 15 modality combination scores. The paired t-test and the Wilcoxon signed rank test shows $p < 0.001$ for both paired comparisons (SSL+HeMIS vs both baselines). These tests are calculated on modality-combination scores for aggregates and not on individual patient level samples, so they are not intended to provide clinical statistical validation, but configuration-level supporting evidence.

D. Class-Wise and Statistical Behavior

In the all-four-modality setting, the model achieves macro precision 0.824, macro recall 0.841 and macro F1-score 0.831. Class-wise F1-scores are 0.784 for NCR, 0.847 for ED, and 0.863 for ET. The confusion matrix analysis of the tumor only gives a correct row-wise classification rate of 77.23% for NCR, 95.32% for ED and 89.89% for ET. This means more reliable segmentation of the ED and ET subregions and the most difficult segmentation of NCR.

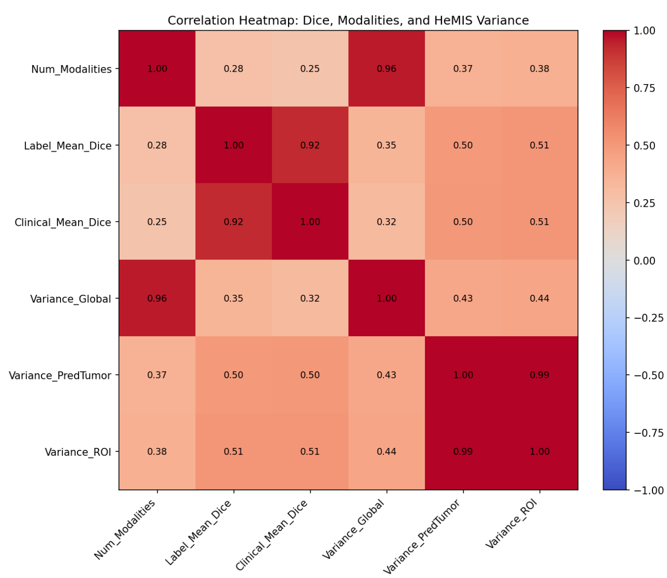

Fig. 5. Correlation heatmap illustrating the association between Dice performance metrics, number of modalities, and HeMIS-based variance indicators.

The correlation analysis between label mean Dice and clinical mean Dice shows their significant correlation with Pearson correlation $r = 0.92$. There is a weak to moderate correlation between number of modalities and label mean Dice ($r = 0.28$) and clinical mean Dice ($r = 0.25$). This confirms the hypothesis that other modalities are beneficial, but that it is also the combination of modalities that is crucial. The approximately moderate relationship between ROI-level HeMIS variance and Dice scores ($r \approx 0.51$) indicates that the two are related and can be correlated. The variance in this research is not used as a clinical confidence score for HeMIS, but rather as a feature-disagreement signal.

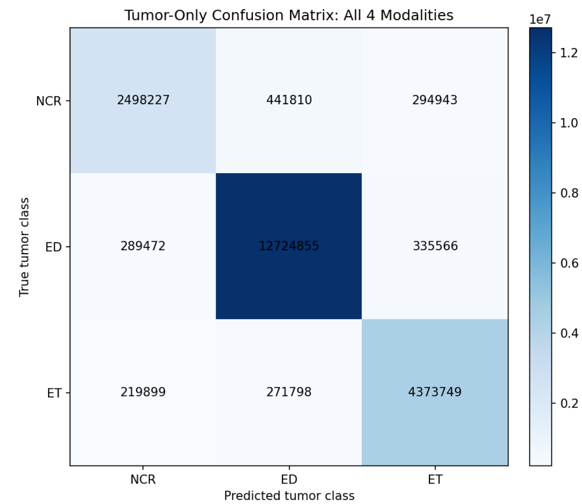

Fig. 6. Tumor-only confusion matrix for all-four-modality prediction. This shows stronger correct classification for ED and ET than for NCR.

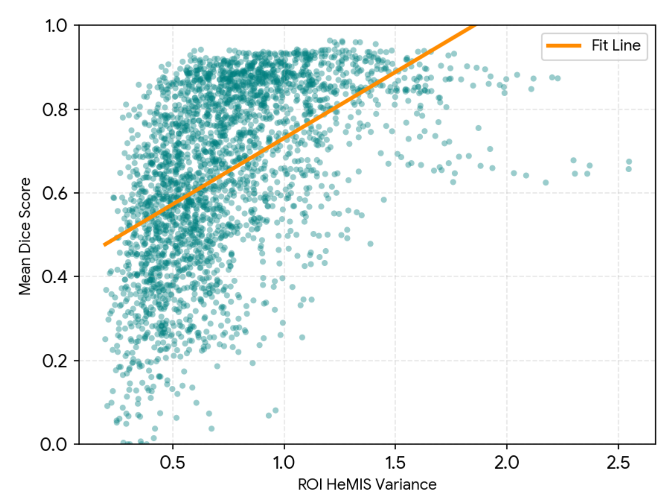

Fig. 7. Per-case ROI variance versus Dice score. HeMIS variance is used as a feature-disagreement signal, not as a calibrated uncertainty score.

E. Qualitative Results

Qualitative segmentation results complement the quantitative results. The predictions with all four modalities agree better with the ground truth mask, and predictions based on a single modality may be incomplete or may not capture any parts of the tumor. A representative case is shown where a complete-input prediction is compared to a FLAIR only prediction in Figure 8. The all-combination visualization in Figure 9 also shows that the model works if there are missing inputs, and that the quality of segmentation is highly dependent on the input modalities available.

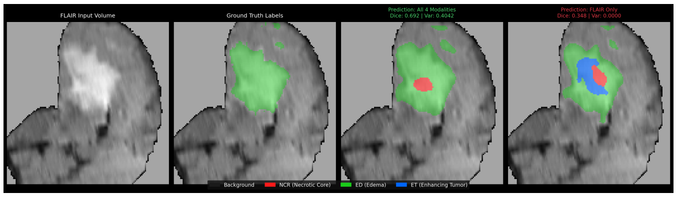

Fig. 8. Representative segmentation output comparing all-four-modality prediction with FLAIR-only prediction.

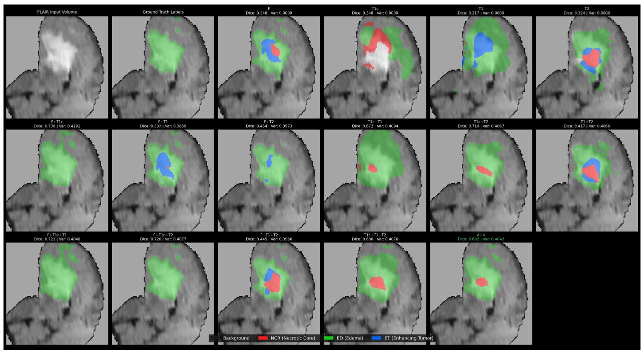

Fig. 9. Representative all-combination qualitative output. Predictions change with modality availability, but the model does not fail completely under incomplete input.

VI. Discussion

The results show that SSL+HeMIS is able to achieve better segmentation performance in the absence of full MRI modalities. All four modalities are optimal for the mode. Concurrently, the model is still applicable in the absence of modalities. This is useful, since the actual MRI procedures may differ in different hospitals, patients and scanners settings. The role of missing-modality fusion can be visualized by comparing the training approach with SimCLR and Med3D. SimCLR adds SSL-based representation learning, while Med3D adds volumetric medical-image modeling, but neither baseline performs modality-aware fusion over available MRI sequences. The suggested framework is a hybrid between the representation learning and HeMIS abstraction, which is more suitable for the missing modality problem. All non-empty subsets of modalities can be processed by the same trained model without training a unique model for each subset. The modally specific behavior is clinically relevant. T1ce is the most useful single modality as it offers contrast enhanced information to aid the identification of enhancing tumor and tumor core. FLAIR and T2 are used for imaging of tumors and edema. However, only the anatomical contrast is not sufficient for the tumor subregion information, and in this case, T1 alone performs poorly.Therefore, the results should not be interpreted as evidence that modalities can be intentionally omitted.

Instead, the model can be considered as a robustness-based segmentation support system for situations in which full imaging is not available.The HeMIS variance analysis should also be interpreted carefully. The variance represents differences between the available modality feature maps. It can be useful to examine representation behavior when using different combinations of modalities. But it is specifically excluded as a calibrated confidence or uncertainty level, and rather as a feature-disagreement signal. Expert review, calibration studies and validation on independent real world data would be required for clinical use.

A. Final Design Decisions

The model was kept for the following reason: It had the best compromise between complete-input accuracy and incomplete-input robustness. The same protocol used for the first evaluation was retained with all 15 non-empty sets of modalities. The final design is also free from the clinical uncertainty score produced by the feature-disagreement analysis, avoiding overclaims.

TABLE VII
SUMMARY OF FINAL DESIGN ADJUSTMENTS

Design aspect	Final adjustment	Reason
Model choice	SSL+HeMIS retained	Strongest robustness across complete and incomplete settings
Evaluation	All 15 non-empty combinations	Matches missing-modality objective
Missing input	Zero-filled SSL input and HeMIS over available modalities	Allows one model to handle different combinations
Metric	Dice-based evaluation	Better for imbalanced tumor segmentation
Variance	Feature-disagreement signal	Avoids uncalibrated clinical uncertainty claims

B. Practical and Ethical Scope

The value of the model is that it can generate segmentation results for a hospital even if the hospital lacks a full set of MRI sequences. This is particularly important in low resource settings, where it may be challenging to get a full 4 modality scan due to cost, scanner time, patient health, and/or protocol variation. However, this is not a substitute for the radiologist but rather as a helper for the radiologist. There is no clinical use unless reviewed and approved by human-in-the-loop. The model is not intended to eliminate the failure to perform clinically useful MR sequences, but rather to decrease failure rates when data are incomplete.

The research is useful for healthcare equity and workflow efficiency. A model trained on unlabeled data can decrease reliance on costly expert annotations, and a lack of dependency on modality can enable hospitals to leverage existing scans. Incomplete scans can be analyzed with a robust analysis to minimize repeated imaging, preparation of the same data and unused historical imaging archives. These claims are in a conditional state and would require a hospital level validation before they can be considered as direct operational benefit.

VII. Practical Impact and Responsibility

This research takes into account social, environmental, ethical and economic issues. These are not experimental claims but are significant for placing the system in the right position.

The social motivation is that many hospitals, particularly in developing areas, may not be able to have complete or labelled imaging data. A model which is able to learn from unlabeled volumes and process missing modalities could make better use of existing MRI archives. This can help to encourage research participation outside of large centers with complete acquisition protocols and fully labeled datasets.

The potential workflow advantage is a quicker initial segmentation. Manual segmentation of brain tumors can be time consuming, and an automatic or semi-automatic mask may assist the radiologist's review of a tumor. Final diagnosis, treatment plan, surgery decision, radiotherapy planning and follow-up interpretation shall always be in the hands of qualified medical experts.

The environmental and economical discussion is also qualitative. With strong models, some repeated scans or unnecessary data preparation could be eliminated if insufficient but clinically satisfactory scans are obtained. Learning from unlabeled data could lessen some of the burden of the experts. These benefits are not guaranteed and did not undergo testing.

Ethically, the model should not be used as a self-contained diagnostic tool. Wrong segmentation may overestimate or underestimate a tumor region, particularly when a modality is missing from the data. In this case, human-in-the-loop would be the responsible deployment model, with clearly defined limitations, scope of the datasets, preprocessing, model behavior, and evaluation protocol documented.

VIII. Limitations and Future Work

The research is limited to the BraTS 2021 dataset. This is a large and standardized data set, but it is curated, and can not be used instead of external clinical validation. In reality, scanner differences, motion artifacts, incomplete acquisition protocols, appearances of rare tumors, and demographic variation can all contribute to real hospital data. Hence, it is not claiming direct clinical use.

One other limitation is that the evaluation is based on held out Fold 0 instead of full 5 fold validation. The generalization over the entire process would have been more well supported if full cross-validation would have been performed. Evaluation can also be performed on newer or external datasets, e.g. later releases of BraTS, to test the robustness of cross-datasets. One direction to explore in the future is to further improve SSL pretraining by increasing the number of pretraining epochs, using a deeper Transformer encoder, a larger embedding dimension, and introducing more attention heads.

The model can also be used to extend to being multi-tasked. The current system generates segmentation masks and doesn't grade the tumor. For future work, the global SSL encoder features could be used for a classification head, keeping segmentation output. Lastly, uncertainty analysis should be codified via caliber methods prior to a clinical belief translation being drawn.

IX. Conclusion

The framework proposed here integrates the masked image modelling self-supervised pre-training with the HeMIS modality fusion. It is trained and evaluated on BraTS 2021 with modalities FLAIR, T1ce, T1, T2, and tested in all 15 combinations without emptiness.

The best segmentation performance is achieved with complete multimodal input with all-four-modality clinical region Dice scores of 0.902 for WT, 0.859 for TC and 0.791 for ET. The model is also valid for the missing modality condition. In each modality combination, SSL+HeMIS outperforms the SSL-SimCLR and SSL-Med3D baselines, highlighting the benefit of using self-supervised MRI representation learning with modality-aware fusion. The framework is a contributing factor to a strong multimodal brain tumor segmentation, however further external validation, uncertainty calibration and expert-in-the-loop evaluation are needed for clinical use.

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